🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3C

Convert Measurement Units

Lesson 3-10 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes (Level 2 Standard)

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can convert measurement units using ratios and conversion factors.
Language Objective: I can explain my conversion using the words conversion factor, unit, equivalent, and ratio.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Convert Measurement Units | Conversion Factor | Unit | Equivalent | Ratio | Customary units | Metric units | Ratio table

👆 *Tap any word to see what it means and a picture.*

1. Renaming a measurement in a different unit without changing the amount is to ________ it.

2. A ratio equal to 1 that relates two units, like 12 inches / 1 foot, is a ________.

3. A standard amount used to measure, like a foot or a gram, is a ________.

4. Two measurements that name the same amount, like 1 foot and 12 inches, are ________.

5. A comparison of two quantities is a ________.

6. Units like inches, feet, and pounds used mainly in the U.S. are

   ____________________ .

7. Units like meters, grams, and liters based on tens are

   ____________________ .

8. A table that lists equivalent ratios in rows or columns is a

   ____________________ .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

JUSTIFY

Will the cabinet fit through the doorway? Show your conversion to prove it?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Convert Measurement Units** — Rename a measurement in a different unit without changing its amount.
  Convertir unidades de medida — Expresar una medida en una unidad diferente sin cambiar su cantidad.
- [ ] **Conversion Factor** — A ratio that helps you change one unit into another.
  Factor de conversión — Una razón que te ayuda a cambiar una unidad por otra.
- [ ] **Unit** — A standard amount used to measure, like an inch or a liter.
  Unidad — Una cantidad estándar para medir, como una pulgada o un litro.
- [ ] **Equivalent** — Having the same value, just in different units.
  Equivalente — Tener el mismo valor, solo en unidades distintas.
- [ ] **Ratio** — A way to compare two amounts, like 3 to 2.
  Razón — Una manera de comparar dos cantidades, como 3 a 2.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

7 feet = 7 × ____ = ____ inches. Since ____ inches is ____ than 80 inches, the cabinet ____ fit through the doorway.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: A foot is bigger than an inch, so I must convert to compare or combine the measurements.

Evidence: Students explain a foot is the larger customary unit (12 inches) and that mixed units must be converted to the same unit before working with them.

Reasoning: This evidence connects the definition of convert measurement units to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **7 feet = 7 × ____ = ____ inches. Since ____ inches is ____ than 80 inches, the cabinet ____ fit through the doorway.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Convert Measurement Units
2. **Notes 2:** Conversion Factor
3. **Notes 3:** Unit
4. **Notes 4:** Equivalent
5. **Notes 5:** Ratio
6. **Notes 6:** Customary units
7. **Notes 7:** Metric units
8. **Notes 8:** Ratio table
9. **Watch for this mistake:** *A common mistake in Convert Measurement Units is applying the conversion factor in the wrong direction — for example, converting 4 feet to inches by dividing (4 ÷ 12 = 0.33 inches) instead of multiplying (4 × 12 = 48 inches), or converting 80 ounces to pounds by multiplying (80 × 16 = 1,280 pounds) instead of dividing (80 ÷ 16 = 5 pounds). Before you answer, ask: am I changing to a SMALLER unit (there will be MORE of them, so multiply) or a BIGGER unit (there will be FEWER of them, so divide)?*
10. **Try It 1:** A. 80 ounces — *Ounces are smaller than pounds, so multiply: 5 × 16 = 80 ounces.*
11. **Try It 2:** A. 2 gallons — *Gallons are bigger than quarts, so divide: 8 ÷ 4 = 2 gallons.*